

Simplifying Expressions with Square Roots (Part I)

Finding Square Roots

① $\sqrt{49} = 7$

② $\sqrt{16} = 4$

③ $\sqrt{25} = 5$

④ $\sqrt{81} = 9$

⑤ $\sqrt{4} = 2$

⑥ $\sqrt{100} = 10$

⑦ $\sqrt{64} = 8$

⑧ $\sqrt{9} = 3$

⑨ $\sqrt{1} = 1$

⑩ $\sqrt{36} = 6$

Calculator Necessary.

⑪ $\sqrt{3} \approx 1.73205$

⑫ $\sqrt{15} \approx 3.87298$

⑬ $\sqrt{2} \approx 1.41421$

⑭ $\sqrt{7} \approx 2.64575$

⑮ $\sqrt{97} \approx 9.84886$

⑯ $\sqrt{\frac{9}{25}} = \frac{\sqrt{9}}{\sqrt{25}} = \frac{3}{5}$

⑰ $\sqrt{\frac{1}{4}} = \frac{\sqrt{1}}{\sqrt{4}} = \frac{1}{2}$

⑱ $\sqrt{\frac{36}{49}} = \frac{\sqrt{36}}{\sqrt{49}} = \frac{6}{7}$

⑲ $\sqrt{\frac{1}{16}} = \frac{\sqrt{1}}{\sqrt{16}} = \frac{1}{4}$

⑳ $\sqrt{\frac{81}{4}} = \frac{\sqrt{81}}{\sqrt{4}} = \frac{9}{2}$

㉑ $\sqrt{\frac{100}{9}} = \frac{\sqrt{100}}{\sqrt{9}} = \frac{10}{3}$

Factoring out a Perfect Square

㉒ $\sqrt{12}$ ㉓ $\sqrt{18}$ ㉔ $\sqrt{27}$

$\sqrt{4 \cdot 3}$ $\sqrt{9 \cdot 2}$ $\sqrt{9 \cdot 3}$

$\sqrt{4 \cdot 3}$ $\sqrt{9 \cdot 2}$ $\sqrt{9 \cdot 3}$

$2 \cdot \sqrt{3}$ $3\sqrt{2}$ $3\sqrt{3}$

$2\sqrt{3}$

㉕ $\sqrt{8}$ ㉖ $\sqrt{90}$ ㉗ $\sqrt{28}$ ㉘ $\sqrt{44}$ ㉙ $\sqrt{54}$

$\sqrt{4 \cdot 2}$ $\sqrt{9 \cdot 10}$ $\sqrt{4 \cdot 7}$ $\sqrt{4 \cdot 11}$ $\sqrt{9 \cdot 6}$

$\sqrt{4 \cdot 2}$ $\sqrt{9 \cdot 10}$ $\sqrt{4 \cdot 7}$ $\sqrt{4 \cdot 11}$ $\sqrt{9 \cdot 6}$

$2\sqrt{2}$ $3\sqrt{10}$ $2\sqrt{7}$ $2\sqrt{11}$ $3\sqrt{6}$

㉚ $\sqrt{20}$ ㉛ $\sqrt{40}$ ㉜ $\sqrt{45}$ ㉝ $\sqrt{24}$ ㉞ $\sqrt{63}$

$\sqrt{4 \cdot 5}$ $\sqrt{4 \cdot 10}$ $\sqrt{9 \cdot 5}$ $\sqrt{4 \cdot 6}$ $\sqrt{9 \cdot 7}$

$\sqrt{4 \cdot 5}$ $\sqrt{4 \cdot 10}$ $\sqrt{9 \cdot 5}$ $\sqrt{4 \cdot 6}$ $\sqrt{9 \cdot 7}$

$2\sqrt{5}$ $2\sqrt{10}$ $3\sqrt{5}$ $2\sqrt{6}$ $3\sqrt{7}$

㉟ $\sqrt{99}$ ㊱ $\sqrt{50}$ ㊲ $\sqrt{32}$ ㊳ $\sqrt{125}$ ㊴ $\sqrt{80}$

$\sqrt{9 \cdot 11}$ $\sqrt{25 \cdot 2}$ $\sqrt{16 \cdot 2}$ $\sqrt{25 \cdot 5}$ $\sqrt{16 \cdot 5}$

$\sqrt{9 \cdot 11}$ $\sqrt{25 \cdot 2}$ $\sqrt{16 \cdot 2}$ $\sqrt{25 \cdot 5}$ $\sqrt{16 \cdot 5}$

$3\sqrt{11}$ $5\sqrt{2}$ $4\sqrt{2}$ $5\sqrt{5}$ $4\sqrt{5}$

㊵ $\sqrt{75}$ ㊶ $\sqrt{48}$ ㊷ $\sqrt{96}$ ㊸ $\sqrt{72}$ ㊹ $\sqrt{112}$

$\sqrt{25 \cdot 3}$ $\sqrt{16 \cdot 3}$ $\sqrt{16 \cdot 6}$ $\sqrt{36 \cdot 2}$ $\sqrt{16 \cdot 7}$

$\sqrt{25 \cdot 3}$ $\sqrt{16 \cdot 3}$ $\sqrt{16 \cdot 6}$ $\sqrt{36 \cdot 2}$ $\sqrt{16 \cdot 7}$

$5\sqrt{3}$ $4\sqrt{3}$ $4\sqrt{6}$ $6\sqrt{2}$ $4\sqrt{7}$

$$\begin{array}{ccccc}
 (45) \sqrt{108} & (46) \sqrt{124} & (47) \sqrt{150} & (48) \sqrt{162} & (49) \sqrt{160} \\
 \sqrt{36 \cdot 3} & \sqrt{4 \cdot 31} & \sqrt{25 \cdot 6} & \sqrt{81 \cdot 2} & \sqrt{16 \cdot 10} \\
 \sqrt{36} \sqrt{3} & \sqrt{4} \sqrt{31} & \sqrt{25} \sqrt{6} & \sqrt{81} \sqrt{2} & \sqrt{16} \sqrt{10} \\
 \boxed{6\sqrt{3}} & \boxed{2\sqrt{31}} & \boxed{5\sqrt{6}} & \boxed{9\sqrt{2}} & \boxed{4\sqrt{10}}
 \end{array}$$

$$\begin{array}{ccccc}
 (50) \sqrt{176} & (51) \sqrt{180} & (52) \sqrt{200} & (53) \sqrt{250} & (54) \sqrt{275} \\
 \sqrt{16 \cdot 11} & \sqrt{36 \cdot 5} & \sqrt{100 \cdot 2} & \sqrt{25 \cdot 10} & \sqrt{25 \cdot 11} \\
 \sqrt{16} \sqrt{11} & \sqrt{36} \sqrt{5} & \sqrt{100} \sqrt{2} & \sqrt{25} \sqrt{10} & \sqrt{25} \sqrt{11} \\
 \boxed{4\sqrt{11}} & \boxed{6\sqrt{5}} & \boxed{10\sqrt{2}} & \boxed{5\sqrt{10}} & \boxed{5\sqrt{11}}
 \end{array}$$

$$\begin{array}{ccccc}
 (55) 3\sqrt{44} & (56) -8\sqrt{50} & (57) 2\sqrt{12} & (58) -3\sqrt{20} & \\
 3\sqrt{4 \cdot 11} & -8\sqrt{25 \cdot 2} & 2\sqrt{4 \cdot 3} & -3\sqrt{4 \cdot 5} & \\
 3\sqrt{4} \sqrt{11} & -8\sqrt{25} \sqrt{2} & 2\sqrt{4} \sqrt{3} & -3\sqrt{4} \sqrt{5} & \\
 3 \cdot 2 \cdot \sqrt{11} & -8 \cdot 5 \cdot \sqrt{2} & 2 \cdot 2 \cdot \sqrt{3} & -3 \cdot 2 \cdot \sqrt{5} & \\
 \boxed{6\sqrt{11}} & \boxed{-40\sqrt{2}} & \boxed{4\sqrt{3}} & \boxed{-6\sqrt{5}} &
 \end{array}$$

$$\begin{array}{ccccc}
 (59) 3\sqrt{84} & (60) 4\sqrt{99} & (61) -5\sqrt{40} & (62) 4\sqrt{88} & \\
 3\sqrt{4 \cdot 21} & 4\sqrt{9 \cdot 11} & -5\sqrt{4 \cdot 10} & 4\sqrt{4 \cdot 22} & \\
 3\sqrt{4} \sqrt{21} & 4\sqrt{9} \sqrt{11} & -5\sqrt{4} \sqrt{10} & 4\sqrt{4} \sqrt{22} & \\
 3 \cdot 2 \cdot \sqrt{21} & 4 \cdot 3 \cdot \sqrt{11} & -5 \cdot 2 \cdot \sqrt{10} & 4 \cdot 2 \cdot \sqrt{22} & \\
 \boxed{6\sqrt{21}} & \boxed{12\sqrt{11}} & \boxed{-10\sqrt{10}} & \boxed{8\sqrt{22}} &
 \end{array}$$

Adding and Subtracting Square Roots

$$\begin{array}{ccc}
 (63) 5\sqrt{3} + 7\sqrt{3} & (64) 8\sqrt{10} - 2\sqrt{10} & (65) 3\sqrt{2} + 7\sqrt{2} \\
 \boxed{12\sqrt{3}} & \boxed{6\sqrt{10}} & \boxed{10\sqrt{2}}
 \end{array}$$

$$\begin{array}{ccc}
 (66) 8\sqrt{5} - 3\sqrt{5} & (67) 8\sqrt{7} + 2\sqrt{7} - \sqrt{7} & (68) -54\sqrt{2} + 9\sqrt{2} - 2\sqrt{2} \\
 \boxed{5\sqrt{5}} & \boxed{9\sqrt{7}} & \boxed{-47\sqrt{2}}
 \end{array}$$

$$\begin{array}{cc}
 (69) 8\sqrt{6} - 5\sqrt{6} + 9\sqrt{6} & (70) -3\sqrt{13} - 9\sqrt{13} + 5\sqrt{13} \\
 \boxed{12\sqrt{6}} & \boxed{-7\sqrt{13}}
 \end{array}$$

$$\begin{array}{c}
 (71) -5\sqrt{17} + 4\sqrt{14} + (-3\sqrt{17}) - (-6\sqrt{14}) \\
 \boxed{-8\sqrt{17} + 10\sqrt{14}}
 \end{array}$$

$$\begin{array}{c}
 (72) -5\sqrt{11} - 4\sqrt{11} - (-6\sqrt{6}) + (-5\sqrt{6}) \\
 \boxed{-9\sqrt{11} + \sqrt{6}}
 \end{array}$$

$$\begin{array}{cc}
 (73) 5\sqrt{3} + \sqrt{12} & (74) \sqrt{54} - 3\sqrt{24} + 2\sqrt{6} \\
 5\sqrt{3} + \sqrt{4 \cdot 3} & \sqrt{9 \cdot 6} - 3\sqrt{4 \cdot 6} + 2\sqrt{6} \\
 5\sqrt{3} + \sqrt{4} \sqrt{3} & \sqrt{9} \sqrt{6} - 3\sqrt{4} \sqrt{6} + 2\sqrt{6} \\
 5\sqrt{3} + 2\sqrt{3} & 3\sqrt{6} - 3 \cdot 2 \cdot \sqrt{6} + 2\sqrt{6} \\
 \boxed{7\sqrt{3}} & 3\sqrt{6} - 6\sqrt{6} + 2\sqrt{6} \\
 & \boxed{-\sqrt{6}}
 \end{array}$$

$$(75) 5\sqrt{2} + 2\sqrt{18}$$

$$5\sqrt{2} + 2\sqrt{9 \cdot 2}$$

$$5\sqrt{2} + 2\sqrt{9 \cdot 2}$$

$$5\sqrt{2} + 2 \cdot 3 \cdot \sqrt{2}$$

$$5\sqrt{2} + 6\sqrt{2}$$

$$\boxed{11\sqrt{2}}$$

$$(77) 3\sqrt{18} - 7\sqrt{2} + 6\sqrt{50}$$

$$3\sqrt{9 \cdot 2} - 7\sqrt{2} + 6\sqrt{25 \cdot 2}$$

$$3\sqrt{9 \cdot 2} - 7\sqrt{2} + 6\sqrt{25 \cdot 2}$$

$$3 \cdot 3\sqrt{2} - 7\sqrt{2} + 6 \cdot 5 \cdot \sqrt{2}$$

$$9\sqrt{2} - 7\sqrt{2} + 30\sqrt{2}$$

$$\boxed{32\sqrt{2}}$$

$$(79) -8\sqrt{11} - 7\sqrt{44} + 6\sqrt{99}$$

$$-8\sqrt{11} - 7\sqrt{4 \cdot 11} + 6\sqrt{9 \cdot 11}$$

$$-8\sqrt{11} - 7\sqrt{4 \cdot 11} + 6\sqrt{9 \cdot 11}$$

$$-8\sqrt{11} - 7 \cdot 2 \cdot \sqrt{11} + 6 \cdot 3 \cdot \sqrt{11}$$

$$-8\sqrt{11} - 14\sqrt{11} + 18\sqrt{11}$$

$$\boxed{-4\sqrt{11}}$$

$$(76) 2\sqrt{8} - 3\sqrt{32} + \sqrt{2}$$

$$2\sqrt{4 \cdot 2} - 3\sqrt{16 \cdot 2} + \sqrt{2}$$

$$2\sqrt{4 \cdot 2} - 3\sqrt{16 \cdot 2} + \sqrt{2}$$

$$2 \cdot 2 \cdot \sqrt{2} - 3 \cdot 4 \cdot \sqrt{2} + \sqrt{2}$$

$$4\sqrt{2} - 12\sqrt{2} + \sqrt{2}$$

$$\boxed{-7\sqrt{2}}$$

$$(78) 2\sqrt{3} - 7\sqrt{75} + 8\sqrt{3}$$

$$2\sqrt{3} - 7\sqrt{25 \cdot 3} + 8\sqrt{3}$$

$$2\sqrt{3} - 7\sqrt{25 \cdot 3} + 8\sqrt{3}$$

$$2\sqrt{3} - 7 \cdot 5 \cdot \sqrt{3} + 8\sqrt{3}$$

$$2\sqrt{3} - 35\sqrt{3} + 8\sqrt{3}$$

$$\boxed{-25\sqrt{3}}$$

$$(80) -9\sqrt{20} - 2\sqrt{125} + 3\sqrt{80}$$

$$-9\sqrt{4 \cdot 5} - 2\sqrt{25 \cdot 5} + 3\sqrt{16 \cdot 5}$$

$$-9\sqrt{4 \cdot 5} - 2\sqrt{25 \cdot 5} + 3\sqrt{16 \cdot 5}$$

$$-9 \cdot 2 \cdot \sqrt{5} - 2 \cdot 5 \cdot \sqrt{5} + 3 \cdot 4 \cdot \sqrt{5}$$

$$-18\sqrt{5} - 10\sqrt{5} + 12\sqrt{5}$$

$$\boxed{-16\sqrt{5}}$$

$$(81) 3\sqrt{80} + (-10\sqrt{20}) - 6\sqrt{45}$$

$$3\sqrt{16 \cdot 5} + (-10\sqrt{4 \cdot 5}) - 6\sqrt{9 \cdot 5}$$

$$3\sqrt{16 \cdot 5} + (-10\sqrt{4 \cdot 5}) - 6\sqrt{9 \cdot 5}$$

$$3 \cdot 4 \cdot \sqrt{5} + (-10 \cdot 2 \cdot \sqrt{5}) - 6 \cdot 3 \cdot \sqrt{5}$$

$$12\sqrt{5} - 20\sqrt{5} - 18\sqrt{5}$$

$$\boxed{-26\sqrt{5}}$$

$$(83) -3\sqrt{28} - 7\sqrt{63} - 6\sqrt{112}$$

$$-3\sqrt{4 \cdot 7} - 7\sqrt{9 \cdot 7} - 6\sqrt{16 \cdot 7}$$

$$-3\sqrt{4 \cdot 7} - 7\sqrt{9 \cdot 7} - 6\sqrt{16 \cdot 7}$$

$$-3 \cdot 2 \cdot \sqrt{7} - 7 \cdot 3 \cdot \sqrt{7} - 6 \cdot 4 \cdot \sqrt{7}$$

$$-6\sqrt{7} - 21\sqrt{7} - 24\sqrt{7}$$

$$\boxed{-51\sqrt{7}}$$

$$(85) 9\sqrt{45} + \sqrt{7} - 2\sqrt{5} + 2\sqrt{28}$$

$$9\sqrt{9 \cdot 5} + \sqrt{7} - 2\sqrt{5} + 2\sqrt{4 \cdot 7}$$

$$9\sqrt{9 \cdot 5} + \sqrt{7} - 2\sqrt{5} + 2\sqrt{4 \cdot 7}$$

$$9 \cdot 3 \cdot \sqrt{5} + \sqrt{7} - 2\sqrt{5} + 2 \cdot 2 \cdot \sqrt{7}$$

$$27\sqrt{5} + \sqrt{7} - 2\sqrt{5} + 4\sqrt{7}$$

$$\boxed{25\sqrt{5} + 5\sqrt{7}}$$

$$(82) 3\sqrt{10} + (-10\sqrt{90}) - 6\sqrt{40}$$

$$3\sqrt{10} + (-10\sqrt{9 \cdot 10}) - 6\sqrt{4 \cdot 10}$$

$$3\sqrt{10} + (-10\sqrt{9 \cdot 10}) - 6\sqrt{4 \cdot 10}$$

$$3\sqrt{10} + (-10 \cdot 3 \cdot \sqrt{10}) - 6 \cdot 2 \cdot \sqrt{10}$$

$$3\sqrt{10} + (-30\sqrt{10}) - 12\sqrt{10}$$

$$\boxed{-39\sqrt{10}}$$

$$(84) -3\sqrt{7} - 4\sqrt{7} - 6\sqrt{28}$$

$$-3\sqrt{7} - 4\sqrt{7} - 6\sqrt{4 \cdot 7}$$

$$-3\sqrt{7} - 4\sqrt{7} - 6\sqrt{4 \cdot 7}$$

$$-3\sqrt{7} - 4\sqrt{7} - 6 \cdot 2 \cdot \sqrt{7}$$

$$-3\sqrt{7} - 4\sqrt{7} - 12\sqrt{7}$$

$$\boxed{-19\sqrt{7}}$$

$$(86) -\sqrt{75} + \sqrt{27} - 6\sqrt{32} + \sqrt{2}$$

$$-\sqrt{25 \cdot 3} + \sqrt{9 \cdot 3} - 6\sqrt{16 \cdot 2} + \sqrt{2}$$

$$-\sqrt{25 \cdot 3} + \sqrt{9 \cdot 3} - 6\sqrt{16 \cdot 2} + \sqrt{2}$$

$$-5\sqrt{3} + 3\sqrt{3} - 6 \cdot 4\sqrt{2} + \sqrt{2}$$

$$-5\sqrt{3} + 3\sqrt{3} - 24\sqrt{2} + \sqrt{2}$$

$$\boxed{-2\sqrt{3} - 23\sqrt{2}}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$$

$1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2, 10^2$

1, 4, 9, 16, 25, 36, 49, 64, 81, 100